

(a) Scope: are physical units held apart?

within-bearing



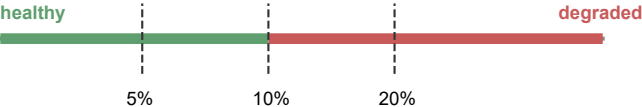
reported false-alarm rate 0.00%

bearing-level holdout



reported false-alarm rate 40.63% (folds 0.4-97.1%)

(c) Definition: where does 'normal' end?



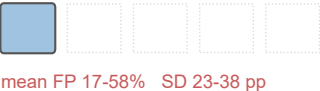
run-to-failure data carry no onset label:
the boundary is chosen by the analyst

SD of the reported rate across five rules

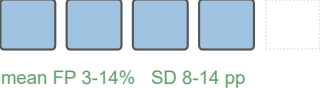
PRONOSTIA (inclusion set fixed): 2.40 pp
XJTU-SY (inclusion set varies): 5.19 pp

(b) Units: how many independent bearings?

1 training bearing (20 recordings)



4 training bearings (same 20 recordings)



3-sigma RMS 6/6 and Isolation Forest 6/6 support it
Mahalanobis saturates at k=1 (67-100% FP): not testable

(d) Reporting checklist

- 1 Report the number of training windows behind the threshold, not the dataset size.
- 2 Report resampling variability from at least two sources: split composition and random seed.
- 3 Report the tie rate when validation resolution is coarser than the false-alarm level of interest.
- 4 Distinguish new recordings from repeated sampling.
- 5 State the number of independent units, and whether held out.